Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Memory Production Paper (6) - My Response to Your Latest Comments

Benjamin D. Summers <bsummers@adagiollc.com> Thu, Sep 24, 2026 at 10:48 PM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>, "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ezequiel and Sara,

Revision 2.1 of the manuscript is on SSRN under the same DOI (https://doi.org/10.2139/ssrn.7206578); the changes are listed below.

Sara,

Thank you for reading the responses. The optimization discussion, Open Problem (v), is conditional as you ask: its sentence on the zero set opens with "For a flat fine family", states that minimizing production and minimizing the charge of the observed family have the same zero set and are two-sidedly comparable by Corollary 4.8, and states that they need not order the cuts identically, since the constants depend on the coarse laws, which depend on the cut. No wording changed there in revision 2.1. The explicit constants you mention are Lemma 2.5 and Proposition 2.6 in Section 2.3 of the revision, two pages; Ezequiel re-derived both and tested them on random families, and Corollary 2.8 sharpens them for process-induced families, so the windows of Table 2 now span factors of 2.2 to 2.8 between lower and upper bound.

Ezequiel,

Thank you for re-deriving the new material and for the five qualifications.

I recomputed every figure in your message. The three windows of Table 2 recompute exactly: 54/456125 with $L^2 = 445/69$, 459/6732200 with 821/69, and 9/28672 for Proposition 6.5. Your constant 110/23 for Example 3.2 and the bound $\sqrt{(9/56375)} = 0.012635$ recompute exactly, and my $h_2 = 0.0265026$ lies inside the sharper window. On 20,000 random triples the worst ratios in Lemma 2.5 are 0.9376 in (a) and 0.9989 in (b), against your 0.938 and 0.9992. On 36 random process-induced families with $n \le 4$ and $m \le 5$ the constants of Proposition 2.6 hold, the sharper constants of the new corollary hold, and the largest value of $h_n/\|\Delta_V\|$ is 0.974, against your 0.976. In Example 4.7 the matrix $LV(2\leftarrow 1)Q_1$ has entries $\pm 1/2$, so its entry size and the total-variation size of each of its columns are 1/2, and its squared norm in the metric (6) is 1; it vanishes on A_1. Your median 7.9% and maximum 63.5% for the source-weighted residual differ from my 5.6% and 50% on 400 process-induced draws by the sampling law, as you say, and the paper now prints an exact instance in place of a range.

Qualification 1, Example 5.4. Every citation of the example now carries the qualification: the two mentions in the Introduction, Remark 2.7, Remark 5.3, the conclusion, and Open Problem (vii) say that the family carries laws and is not process-induced. The restricted question is Open Problem (vii): can a process-induced family have more than one nearest flat family? I do not know the answer. On two symmetric slices, where non-uniqueness would appear as a pair of minimizers exchanged by the symmetry, the nearest flat family was unique in every case I computed: the two-state families (K_a, K_a, K_c) with K_a the symmetric kernel with diagonal $a \ge 1/2$, over the realizable range $2a - 1 \le c \le 1$, and the three-state reversal-symmetric families $(A, A^{\mathsf{T}}, A^{\mathsf{T}}A + x\,\iota_0[\delta_0])$, with A and $\iota_0[\delta_0]$ from the interference construction, over the range of x on which the joint law is non-negative. At the two boundary points (a, c) = (1/2, 0) and (1/2, 1) the minimizer (J/2, J/2) is unique and degenerate to second order, with the squared distance $1 + 16x^4$ along one line through it. This is evidence and not an answer, and the paper prints the question only.

Qualification 2, the constants. The argument delivers the sharper lower bound for every family that transports its laws, and Corollary 2.8 states it: for $V(j\leftarrow i)p_i = p_j$ at all pairs, $\|\Delta_V\|/(2 + 1/\min_v p_2(v))^{\{1/2\}} \le h_n(V) \le \|\Delta_V\|$ on the three-point grid, and on m + 1 points the lower constant is $((m-1)(2 + 1/\rho))^{\{1/2\}}$ and the upper constant is $(m(m-1)/2)^{\{1/2\}}$, since with the one-step kernels transporting the laws every multiplication in the telescoping is a contraction. Example 5.4 is not a witness for the general statement: I and σ preserve the uniform law, so that family transports its laws; it is not process-induced because no joint law has its three margins. Table 2 uses the corollary, all four families being process-induced: $\sqrt{(9/56375)} \le h_2 \le \sqrt{(18/23575)}$ for Example 3.2, a factor 2.19 in place of 2.54; $\sqrt{(51/475600)} \le h_4 \le \sqrt{(153/188600)}$ for Example 3.3, 2.75 in place of 3.45; and $3/(64\sqrt{5}) \le h_3 \le 3/64$ for Proposition 6.5, 2.24 in place of 2.65. On C*: I cannot prove $C^* = 1$ on every grid, and the paper poses it as Open Problem (vi). At every flat family with process-induced laws that I sampled, 175 families with $n \le 4$ and $m \le 4$, the smallest nonzero singular value of the linearized deficit map $E \mapsto \{E_{ts} - W_{tr} E_{rs} - E_{tr} W_{rs}\}$, taken on the orthogonal complement of the tangent space of Flat_n in the metric (6), was at least 1, the smallest value being 1.00003; so at those families, to first order, the ratio $h_n/\|\Delta_V\|$ does not exceed 1, and the constant 1 is approached. Families

placed along the worst linear direction at distance 0.02 to 0.05 from those flat families gave ratios up to 0.999. The upper-bound argument compares the flat family of V's own one-step kernels with the consecutive-cut deficits only, and a proof of C* = 1 would need a different flat family or a different accounting.

Qualification 3, Remark 2.4. The remark now states the source-weighted norm of a lift, $\|\iota_i[\delta]\|^2 = \Sigma_{a,b} (L\omega_i)(a)\, \delta_{ba}^2\, \Sigma_{v\in\lambda^{-1}(b)} p_j(v)^2/p^\lambda_j(b)^2$, whose inner sum is below 1 on every block with two or more states, so no member of the class makes the lift an isometry, and it gives an exact process-induced instance: laws (1/2, 1/4, 1/4) at all three times, $V(1\leftarrow 0) = V(2\leftarrow 1) = p_1^\intercal$, $V(2\leftarrow 0) = p_1^\intercal + (1/8)\, \iota[\delta_1]$ with $\delta_1 = [[1, -3], [-1, 3]]$, joint law $p(u)p(v)V(2\leftarrow 0)_{wu} \geq 1/128$, and the cut $\{1, 2\} \to a$, $\{3\} \to b$. The deficit is the block lift $(1/8)\, \iota[\delta_1]$, so $\Phi = \delta_1/8$ and $\Xi = 0$; in (6), $\|\Delta V\|^2 = \|\Phi\|^2 = 1/4$; with the source weights, $\|\Delta V\|^2 = 7/96$ against $\|\Phi\|^2 = 9/96$, a failure of 1/48. No percentage is printed.

Qualification 4. Remark 2.7 and the conclusion state that at $d = n$ the charge and the deficit norm are comparable, with constants depending only on m and ρ, so the two carry the same information up to those constants and the content of the profile beyond the deficit norm lies at $d > n$. The instance is the coarse family of Example 3.2: $h_2 > 0$, and its profile vanishes at $d = 3$ in the time-inhomogeneous class, since the family is the compression of the three-state chain. I write comparable rather than bi-Lipschitz, since the statement bounds the two functions against each other and says nothing about their moduli of continuity.

Qualification 5. Remark 5.3 cites Erdős's theorem for the null set (Bull. Amer. Math. Soc. 52, 1946, Theorem 1): for a closed subset of a Euclidean space the points with more than one nearest point lie in countably many surfaces of finite measure of codimension one. It cites nothing else for it.

Your outstanding check: Remark 2.7 refers to Proposition 5.3 of the projection paper, the Łojasiewicz comparison, whose number revision 2.1 of that paper preserves; Proposition 2.6 here gives that comparison at $d = n$ with exponent one and explicit constants in the metric (6). Nothing else in this paper depends on that paper's revision.

Numbering: Corollary 2.8 and Open Problems (vi) and (vii) are appended, the corollary's two displays are unnumbered, and every existing result and equation number stands. The reference list gains Erdős at [5], so the references from [5] on move up by one; the projection paper is [22].

Ben

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On Mon, Sep 21, 2026 at 8:55 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:

> Ben,
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> Thank you for your detailed response to Sara's and my independent reports.
> I have compared your response with the new version of the manuscript and re-derived the principal new arguments independently.
> The revision addresses the substantive discrepancies we identified between the statements of the results and their proved scope.
> My assessment of this paper, following this response cycle, is favourable, subject to the qualifications below.
>
> **Points resolved in the revised manuscript**
>
> Section 2.1 now states explicitly that the metric (6) lies outside the companion's source-weighted admissible class, and supplies the equivalence constants; these are immediate, since the ratio of weights is $1/p_j(v)$.
> Lemma 2.5 and Proposition 2.6 give a direct comparison between the dimension-matched charge and the deficit norm in the stated law metric,
> which removes the dependence at $d = n$ altogether.
> The abstract, Corollary 4.8 and the conclusion now distinguish the exact deficit identities from the resulting bounds on the charge,
> and the lower bound of Corollary 4.8, restated as $\|\Phi + P\|/L^*$, is no longer vacuous in the mixed regime.
>
> The revision separates the two-time Chapman–Kolmogorov condition from Markovianity of the observed process.
> Corollary 4.6 is appropriately renamed, and Example 4.7 supplies a valid flat fine chain whose observed bits are pairwise independent
> and whose three-time law is not Markov. The same example distinguishes full dynamical closure from closure on the accessible subspace,